

1) Настройка STP с явным назначением корневого коммутатора

Назначаем Layer2Switch-1 корневым мостом, установив приоритет 4096:

enable

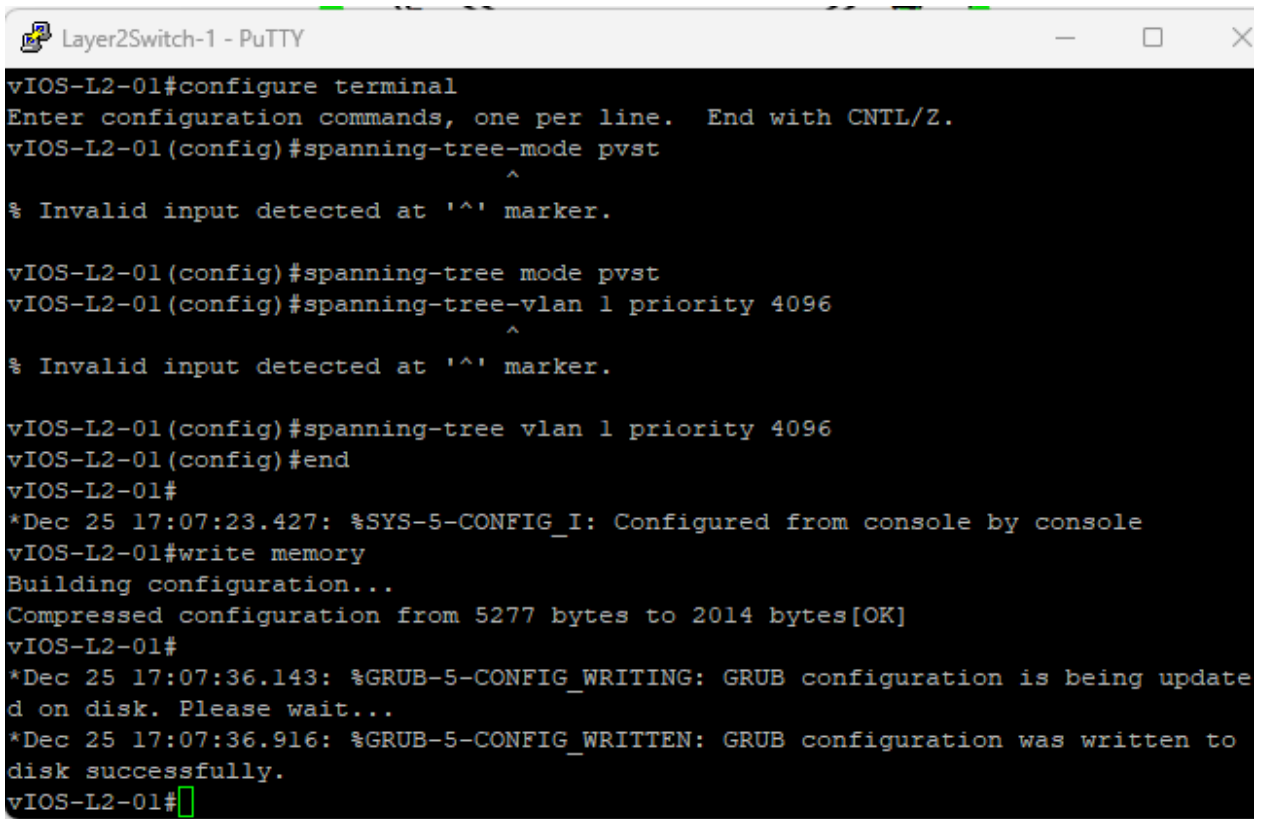
configure terminal

spanning-tree mode pvst

spanning-tree vlan 1 priority 4096

end

write memory



```
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree-mode pvst
^
% Invalid input detected at '^' marker.

vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#spanning-tree-vlan 1 priority 4096
^
% Invalid input detected at '^' marker.

vIOS-L2-01(config)#spanning-tree vlan 1 priority 4096
vIOS-L2-01(config)#end
vIOS-L2-01#
*Dec 25 17:07:23.427: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5277 bytes to 2014 bytes[OK]
vIOS-L2-01#
*Dec 25 17:07:36.143: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Dec 25 17:07:36.916: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

На остальных коммутаторах оставляем приоритет по умолчанию:

enable

configure terminal

spanning-tree mode pvst

end

write memory

```
Layer2Switch-2 - PuTTY

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* conditions of the Cisco End User License Agreement at
* http://www.cisco.com/go/eula
*
* Unauthorized use or distribution of this software is expressly
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*****
vIOS-L2-01>enable
vIOS-L2-01#configure termina
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#end
vIOS-L2-01#w
*Dec 25 17:09:00.577: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5240 bytes to 1993 bytes[OK]
vIOS-L2-01#
*Dec 25 17:09:10.566: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Dec 25 17:09:11.344: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

```
Layer2Switch-3 - PuTTY

*
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*****
vIOS-L2-01>enable
vIOS-L2-01#co
vIOS-L2-01#confi
vIOS-L2-01#configure ter
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spa
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#end
vIOS-L2-01#
*Dec 25 17:10:01.065: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5042 bytes to 1920 bytes[OK]
vIOS-L2-01#
*Dec 25 17:10:11.377: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Dec 25 17:10:12.132: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

```
Layer2Switch-4 - PuTTY
*
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*
*****
vIOS-L2-01>enable
vIOS-L2-01#conf
vIOS-L2-01#configure ter
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spa
vIOS-L2-01(config)#spanning-tree mode pvs
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#end
vIOS-L2-01#
*Dec 25 17:10:50.353: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5042 bytes to 1920 bytes[OK]
vIOS-L2-01#
*Dec 25 17:10:59.222: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Dec 25 17:10:59.983: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

```
Layer2Switch-5 - PuTTY
*
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* Prohibited.
*
*****
vIOS-L2-01>enable
vIOS-L2-01#confi
vIOS-L2-01#configure ter
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spa
vIOS-L2-01(config)#spanning-tree mode pv
vIOS-L2-01(config)#spanning-tree mode pvst
vIOS-L2-01(config)#end
vIOS-L2-01#w
*Dec 25 17:11:39.427: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5042 bytes to 1924 bytes[OK]
vIOS-L2-01#
*Dec 25 17:11:49.217: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Dec 25 17:11:49.976: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

2) Проверка доступности между всеми VPC

Выполняем ring между всеми компьютерами:

PC1 - PuTTY

```
PC1> ip 192.168.1.2/24 192.168.1.1
Checking for duplicate address...
PC1 : 192.168.1.2 255.255.255.0 gateway 192.168.1.1

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=4.541 ms
^C
PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=9.900 ms
^C
PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=9.349 ms
^C
PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=7.461 ms
^C
PC1> ping 192.168.1.7

84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=10.298 ms
^C
PC1> 
```

PC2 - PuTTY

```
PC2> ip 192.168.1.3/24 192.168.1.1
Checking for duplicate address...
PC2 : 192.168.1.3 255.255.255.0 gateway 192.168.1.1

PC2> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=7.177 ms
^C
PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=10.628 ms
^C
PC2> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=3.897 ms
^C
PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=14.146 ms
^C84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=11.937 ms
^[84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=4.543 ms
^X84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=9.652 ms
^Z^C
PC2> ping 192.168.1.7

84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=11.079 ms
^C
PC2> 
```

 PC3 - PuTTY

```
PC3> ip 192.168.1.4/24 192.168.1.1
Checking for duplicate address...
PC3 : 192.168.1.4 255.255.255.0 gateway 192.168.1.1

PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=6.940 ms
^C
PC3> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=8.541 ms
^C
PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=2.400 ms
^C
PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=7.431 ms
^C
PC3> ping 192.168.1.7

84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=10.785 ms
^C
PC3> 
```

 PC4 - PuTTY

```
PC4> ip 192.168.1.5/24 192.168.1.1
Checking for duplicate address...
PC4 : 192.168.1.5 255.255.255.0 gateway 192.168.1.1

PC4> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=8.982 ms
^C
PC4> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=7.993 ms
^C
PC4> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=5.533 ms
^X^C
PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=9.114 ms
^C
PC4> ping 192.168.1.7

84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=8.147 ms
^C
PC4> 
```

PC5 - PuTTY

```
PC5> ip 192.168.1.6/24 192.168.1.1
Checking for duplicate address...
PC5 : 192.168.1.6 255.255.255.0 gateway 192.168.1.1

PC5> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=6.564 ms
^C
PC5> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=6.049 ms
^C
PC5> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=6.030 ms
^C
PC5> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=6.269 ms
^C
PC5> ping 192.168.1.7

84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=5.107 ms
^C
PC5> 
```

PC6 - PuTTY

```
PC6> ip 192.168.1.7/24 192.168.1.1
Checking for duplicate address...
PC6 : 192.168.1.7 255.255.255.0 gateway 192.168.1.1

PC6> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=10.530 ms
^C
PC6> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=9.451 ms
^C
PC6> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=6.899 ms
^C
PC6> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=9.666 ms
^C
PC6> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=8.270 ms
^C
PC6> 
```

PC1> ping 192.168.1.2 # к PC2

PC1> ping 192.168.1.3 # к PC3

PC1> ping 192.168.1.4 # к PC4

PC1> ping 192.168.1.5 # к PC5

PC1> ping 192.168.1.6 # к PC6

Аналогично с остальных PC...

Результаты: Все ping должны быть успешными, так как STP заблокировал только избыточные пути, но сеть осталась связной.

3) Сбор информации о BID, портах и стоимостях

Команды для сбора информации:

На каждом коммутаторе

show spanning-tree

```
vIOS-L2-01>show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID      Priority    4097
```

```
Address      0c19.6477.0000
```

```
This bridge is the root
```

```
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Bridge ID    Priority    4097  (priority 4096 sys-id-ext 1)
```

```
Address      0c19.6477.0000
```

```
Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

```
Aging Time   300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	Shr
Gi0/1	Desg	FWD	4	128.2	Shr
Gi0/2	Desg	FWD	4	128.3	Shr
Gi0/3	Desg	FWD	4	128.4	Shr
Gil/0	Desg	FWD	4	128.5	Shr
Gil/1	Desg	FWD	4	128.6	Shr
Gil/2	Desg	FWD	4	128.7	Shr
Gil/3	Desg	FWD	4	128.8	Shr

```
--More--
```

```
Layer2Switch-2 - PuTTY

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID      Priority    4097
              Address     0c19.6477.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
              Address     0c94.3477.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Root FWD 4        128.1   Shr
Gi0/1          Altn BLK 4        128.2   Shr
Gi0/2          Desg FWD 4        128.3   Shr
Gi0/3          Desg FWD 4        128.4   Shr
Gi1/0          Desg FWD 4        128.5   Shr
Gi1/1          Desg FWD 4        128.6   Shr
Gi1/2          Altn BLK 4        128.7   Shr
--More--
```

```
Layer2Switch-3 - PuTTY

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* http://www.cisco.com/go/eula
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*****
vIOS-L2-01>show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID      Priority    4097
              Address     0c19.6477.0000
              Cost        4
              Port        1 (GigabitEthernet0/0)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID    Priority    32769 (priority 32768 sys-id-ext 1)
              Address     0cce.724e.0000
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time   300 sec

Interface      Role Sts Cost      Prio.Nbr Type
-----
Gi0/0          Root FWD 4        128.1   Shr
Gi0/1          Altn BLK 4        128.2   Shr
Gi0/2          Altn BLK 4        128.3   Shr
Gi0/3          Altn BLK 4        128.4   Shr
Gi1/0          Desg FWD 4        128.5   Shr
Gi1/1          Desg FWD 4        128.6   Shr
--More--
```

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vIOS-L2-01>show spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 4097
 Address 0c19.6477.0000
 Cost 4
 Port 1 (GigabitEthernet0/0)
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
 Address 0cf7.b14b.0000
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	Shr
Gi0/1	Altn	BLK	4	128.2	Shr
Gi0/2	Altn	BLK	4	128.3	Shr
Gi0/3	Altn	BLK	4	128.4	Shr
Gil/0	Desg	FWD	4	128.5	Shr
Gil/1	Desg	FWD	4	128.6	Shr

--More--

```
Layer2Switch-5 - PuTTY
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* http://www.cisco.com/go/eula
*
* Unauthorized use or distribution of this software is expressly
* Prohibited.
*****
vIOS-L2-01>show spanning-tree

VLAN0001

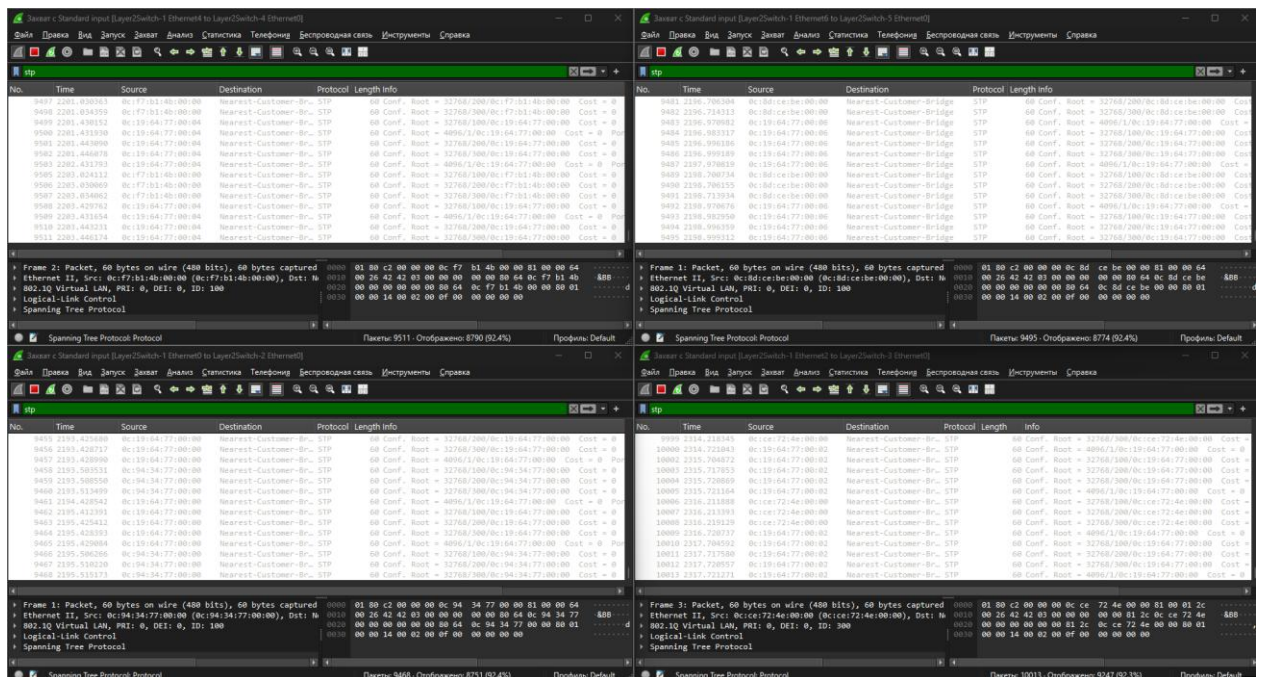
Spanning tree enabled protocol ieee
Root ID    Priority    4097
Address    0c19.6477.0000
Cost       4
Port       1 (GigabitEthernet0/0)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
Address     0c8d.cebe.0000
Hello Time  2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time  300 sec

Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/0 Root FWD 4 128.1 Shr
Gi0/1 Altn BLK 4 128.2 Shr
Gi0/2 Desg FWD 4 128.3 Shr
Gi0/3 Desg FWD 4 128.4 Shr
Gi1/0 Desg FWD 4 128.5 Shr
Gi1/1 Desg FWD 4 128.6 Shr

--More--
```

4) Отслеживание Hello-пакетов в Wireshark



Команды для настройки перехвата:

Результаты:

Hello-пакеты отправляются каждые 2 секунды (по умолчанию в STP)

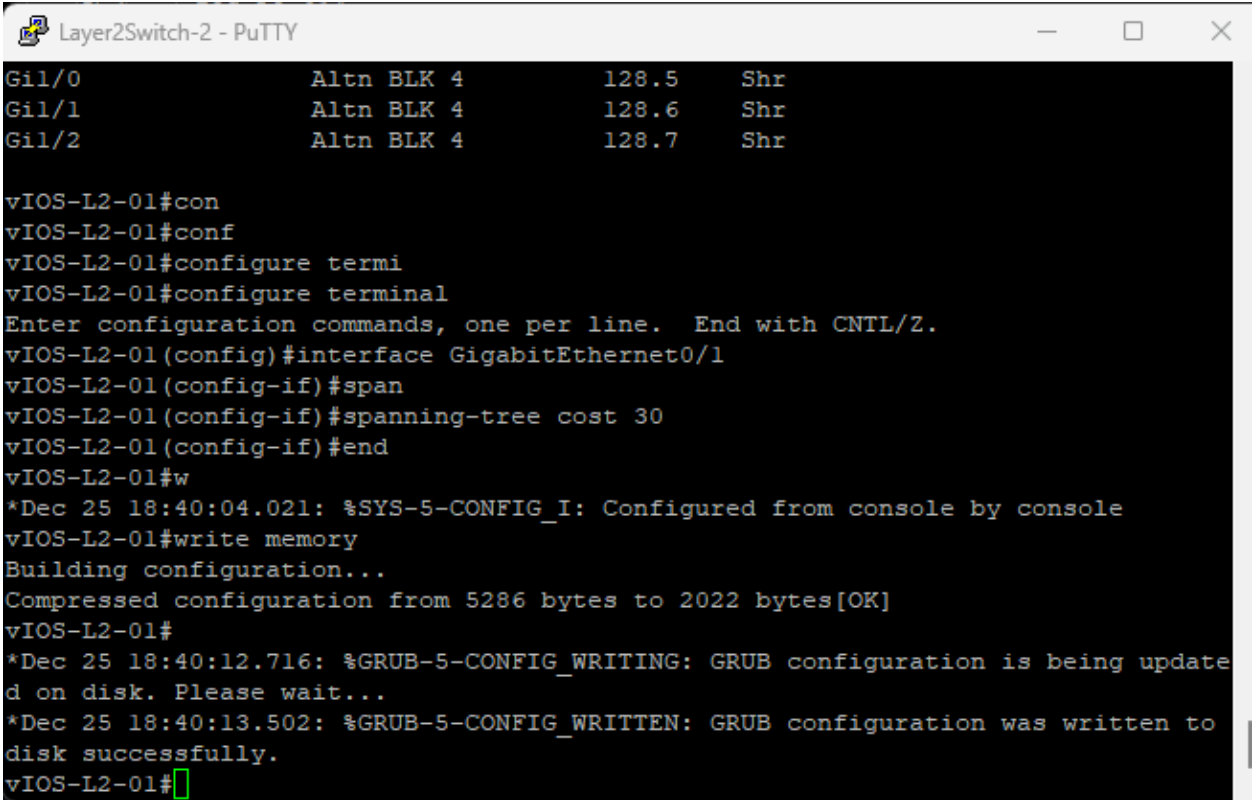
Источник: корневой коммутатор (Layer2Switch-1)

На заблокированных портах BPDU принимаются, но не пересылаются

Максимальное время жизни (Max Age) — 20 секунд

5) Изменение стоимости маршрута

Выбираем порт на назначенном коммутаторе (например, Fa0/1 на Layer2Switch-2):



```
Layer2Switch-2 - PuTTY
Gi1/0          Altn BLK 4          128.5      Shr
Gi1/1          Altn BLK 4          128.6      Shr
Gi1/2          Altn BLK 4          128.7      Shr

vIOS-L2-01#con
vIOS-L2-01#conf
vIOS-L2-01#configure termi
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interface GigabitEthernet0/1
vIOS-L2-01(config-if)#span
vIOS-L2-01(config-if)#spanning-tree cost 30
vIOS-L2-01(config-if)#end
vIOS-L2-01#w
*Dec 25 18:40:04.021: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5286 bytes to 2022 bytes[OK]
vIOS-L2-01#
*Dec 25 18:40:12.716: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*Dec 25 18:40:13.502: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

configure terminal

interface FastEthernet0/1

spanning-tree cost 30

end

write memory

Повторяем команду show spanning-tree и сохраняем изменения на схеме.

```
Layer2Switch-2 - PuTTY

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    4097
            Address    0c61.e63e.0000
            Cost        8
            Port        5 (GigabitEthernet1/0)
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address    0cba.bed3.0000
            Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time   15 sec

Interface                Role Sts Cost      Prio.Nbr Type
-----
Gi0/0                    Altn BLK 30      128.1 Shr
Gi0/1                    Altn BLK 30      128.2 Shr
Gi0/2                    Altn BLK 4      128.3 Shr
Gi0/3                    Altn BLK 4      128.4 Shr
Gi1/0                    Root FWD 4      128.5 Shr
Gi1/1                    Altn BLK 4      128.6 Shr
Gi1/2                    Altn BLK 4      128.7 Shr
--More--
```

6) Сохранение конфигураций

На каждом устройстве

copy running-config